Guess the Number

Project Overview

Guess the Number is a Python-based number guessing game where the player attempts to guess a randomly generated number within a specified range. The game offers three difficulty levels, provides hints after incorrect guesses, and tracks player statistics across games.

Submission Contents

- Dangould_Python_Game_Code.py
- DanGould_Python_Game_Flowchart.pdf
- DanGoud_Python_Game_Presentation.mp4

Repository

The source code and project documentation are available on Github:

https://github.com/Dan-Gould/guess-the-number.git

Features

- **Three difficulty levels:** Easy, Medium and Hard, each with a different number range and attempt limit.
- **Random number generation**: A new target number is generated for each game.
- **Input validation:** Checks that guesses are valid integers within the permitted range and that difficulty and menu selections are valid.
- **Higher and lower hints:** Helps the player narrow down the target number after an incorrect guess.
- **Odd/even clue:** Provides an additional clue before the final attempt.
- **Game statistics:** Tracks games played, games won, games lost and win rate.
- **Best game tracking:** Records the fewest attempts needed to win, favouring the higher difficulty when the number of attempts is tied.
- **Post-game menu:** Allows the player to start another game, view statistics, revisit the instructions or exit.

Difficulty Levels

Difficulty	Number Range	Maximum Attempts
Easy	1-50	7
Medium	1-100	6
Hard	1-200	5

How to Play

1. Run the Python program.
2. Read the instructions displayed at the start.
3. Choose a difficulty level: Easy, Medium or Hard.
4. Enter a number within the displayed range.
5. Follow the hints to determine whether your guess is too high or too low.
6. Use the odd/even clue when it appears.
7. Guess the target number before running out of attempts to win.
8. After the game, use the menu to play again, view statistics, see the instructions or exit.

Invalid guesses and menu selections are rejected, allowing the player to enter another response.